

String functions:

Convert to lowercase:

```
df["name"] = df["name"].str.lower()
```

```
df["name"] = df["name"].str.upper()
```

```
df["name"] = df["name"].str.capitalize()
```

Replace text:

```
df["city"] = df["city"].str.replace("madras",  
"chennai")
```

check if string contains word: $\Rightarrow$ `isin()`, `text`, `NLP`

```
df[df["review"].str.contains("good")]
```

Where use:

check if string contains a word

NLP, sentiment, text.

Count

```
df.groupby("city").count()
```

groupby

groups rows having the same value.

Max:

```
df.groupby("city")["salary"].max
```

Minimum:

df.groupby("city")["salary"].min

Aggregate functions:

Built in function in pandas

sum()

mean()

count()

Max()

Min()

Median()

std()

Merging dataframes: → 2 dataframes

students:

id	name
1	John
2	David

→ 1 dataframe

Marks:

id	Marks
1	90
2	85

→ 2 dataframes

Output:

id	name	Marks
1	John	90
2	David	85

merge()

pd.merge(df1, df2, on='id') syntax:

Concatenation: → 2 dataframes with same columns

pd.concat([df1, df2])

Customer Review dataset:

1. checking the Missing Values
2. Fill Missing Values
3. Removing duplicates
4. Convert Reviews into lowercase.
5. group by sentiment.

Ques:

~~1. Convert the name column to uppercase?~~

1. Convert the name column to uppercase?

2. Replace "madras" with "chennai"?

3. Find the Average Salary for each city with groupby()?

4. Merge two dataframes using the id column:

— x —

Advanced Pandas for AI and Gen AI

apply() $\Rightarrow$ use Series & dataframe

* Is used to apply a function to every value in a Series or along rows and columns of a dataframe

* Instead of writing loops, we can use apply()

```
import pandas as pd
```

```
dt = pd.DataFrame({"marks": [80, 90, 70, 60]})
```

```
dt["marks"] = dt["marks"].apply(lambda x: x+5)
```

```
print(dt)
```

Output:

	Marks
0	85
1	95
2	75
3	65

unnamed function.
small one line function

lambda function:

It's a small one line function without a function name

Syntax:

lambda arguments:

Arguments - Input values
Expression - Calculations performed and
Returned the expression automatically.

```
Sq = lambda x: x * x  
print(square(5))  
output:  
25
```

Map()

Series

```
df["gender"] = df["gender"].map({"m":  
    "male", "f": "female"})
```

Output:

Male

Female

Male

Complete AP function:

[common flow]

end to end AP flow:

Customer - Review - dataset

reading the CSV file

explore head, tail, info, describe

handling the missing values

Remove the duplicates

Convert Reviews to lowercase

Filter positive Reviews

Group by product

Save the clean dataset

feed into machine learning; nlp, llm,
rag pipeline

AP pipeline:

collect data → load data with pandas →

clean data → analyse data →

Feature engineering → train machine learning /

deep learning model → evaluate → deploy

- Qn:
1. Create a dataframe with name, marks
increase every mark by 10 using apply()
 2. Convert all Student names to
Upper Case
 3. Create a Salary Column and increase it
by 20%.

Coerce:

valid Numbers stay as Numbers
Invalid becomes NaN

ex:	After
Age	Age
20	20
22	22
twenty	NaN
NA	NaN

Merge two file ex:

Students.csv & Teachers.csv

```
import pandas as pd
Students = pd.read_csv("students.csv")
teachers = pd.read_csv("Teachers.csv")
merge = pd.merge(Students, teachers, on="id")
print(merge)
```

~~Find the average~~
Replace Missing age values with average age?

```
import pandas as pd
df["age"] = df["age"].fillna(df["age"].mean())
print(df)
```

Create a dataframe with name, marks:

```
import pandas as pd
data = {
    "name": ["Elsa", "Aana", "Olat"],
    "age": [95, 99, 88]
}
df = pd.DataFrame(data)
print(df)
```

Create a salary column and increase by 20%:

```
import pandas as pd
data = {
    "name": ["Elsa", "Aana", "Olat"],
    "salary": [30000, 20000, 10000]
}
df = pd.DataFrame(data)
df["salary"] = df["salary"] * 1.20
print(df)
```